

1

Multiple Choice 1 point Practice Question

Which register is used to divide the CPU clock frequency to generate a slower Timer Clock frequency?

- ☐ Auto-Reload Register (ARR)
- ☐ Capture Compare Register (CCR)
- ☒ Prescaler Register (PSC)
- ☐ Control Register (CR1)

2

Multiple Choice 1 point Practice Question

5. In Center-aligned Counting Mode, the counter counts:

- ☐ Up from 0 to ARR, then resets to 0.B.
- ☐ Down from ARR to 0, then resets to ARR.
- ☒ Up from 0 to ARR, then Down from ARR to 0.
- ☐ Randomly between 0 and ARR.

Assume a 4-bit system. For each of the following operations, compute the result and NZCV flags, based on the first row that has been given to you. (If the result is incorrect, write the correct result in parenthesis like (true = 9).)

Binary Arithmetic

Operation	Result in binary	Equivalent unsigned arithmetic in decimal	Equivalent signed arithmetic in decimal	NZCV
0100 + 0101	1001	$4 + 5 = 9$	$4 + 5 = -7$ (true = 9)	1001
1001 + 0010	1011	$9 + 2 = 11$	$-7 + 2 = -5$	1000
0111 + 0110	1101	$7 + 6 = 13$	$7 + 6 = -3$ (true = 13)	1101
1010 - 1010	0000	$10 - 10 = 0$	$-6 - (-6) = 0$	0100
1001 + 1001	0010	$9 + 9 = 2$ (true = 18)	$-7 + -7 = 2$ (true = -14)	0011

Given R0 = 0x81223344, write its value after executing each instruction (individually, not sequentially).

1) MOV R1, R0, ASR #4

R1 = 0x

2) MOV R2, R0, ASR #8

R2 = 0x

3) MOV R3, R0, ASR #16

R3 = 0x

4) MOV R4, R0, LSR #8

R4 = 0x

5) MOV R5, R0, LSR #31

R5 = 0x

Find the register values (in decimal) after finishing execution of the following program (sequentially, not individually).

```
MOV R6, #4  
MOV R8, #3  
MOV R6, R6, LSL  
#5  
ADD R9, R8, R8,  
LSL #2  
RSB R10, R9, LSL  
#3  
ADD R7, R6, R9  
SUB R11, R10, R8
```

Output:

R6 = 128

R9 = 15

R10 = 105

R7 = 143

R11 = 102

Consider the following C program. Fill in the blanks for the corresponding assembly programs (two versions).

C Program	Assembly Program 1 (using branch statement)	Assembly Program 2 (using conditional execution)
<pre>if ((r1 == r3) && (r5 == r6)) r7 = r7 + 10</pre>	<pre>CMP r1,r3 BNE SKIP CMP r5,r6 BNE SKIP ADD r7,r7,#10 SKIP:</pre>	<pre>CMP r1,r3 CMPEQ r5,r6 ADDEQ r7,r7,#10</pre>

Consider the following C program that swaps the values of two chars. Fill in the blanks for the corresponding assembly program.

C Program	Assembly Program
<pre>void swap (char *x, char *y) { char t; t = *x; *x = *y; *y = t; }</pre>	<pre>swap PROC EXPORT swap LDRB r2,[r0] ; Load a byte, r0 = *x LDRB r3,[r1] ; Load a byte, r1 = *y STRB r2,[r1] STRB r3,[r0] BX LR ENDP</pre>

Suppose a 16-bit timer has the following settings:

1. CPU clock frequency = 72 MHz.
2. The prescaler PSC = 71
3. ARR = 999
4. CCR = 250
5. Counting direction: down-counting
6. Output is set as PWM mode 1

Calculate Timer clock frequency = MHz; Timer Frequency
= KHz; PWM duty cycle = .

Show all updates to registers as the assembly code shown below runs, assuming **little-endian** ordering. Fill in the tables of final register values and memory contents. Show the NZCV flags after execution, assuming NZCV=0000 initially.

```
LDR R1, #0x10000010
LDR R2, [R1]
LDR R3, [R1, #4]
MOV R4, R3, ROR #16
MOV R5, 0x00000F00
LSLS R6, R5, #8
BIC R7, R2, R6
AND R8, R7, R6
STR R7, [R1, #8]
```

Initial memory content at 0x10000010 contains the following bytes in increasing memory address order:

0x10000010 FF EF CD AB 00 00 CD AB AA BB CC DD EE FF 11 22

Final register values:

R1 = 0x 10000010

R2 = 0x ABCDEFFF

R3 = 0x ABCD0000

R4 = 0x 0000ABCD

R5 = 0x 00000F00

R6 = 0x 000F0000

R7 = 0x ABC0EFFF

R8 = 0x 00000000

Final memory contents:

0x10000010 FF EF CD AB 00 00 CD AB EE FF 11 22

Final NZCV =